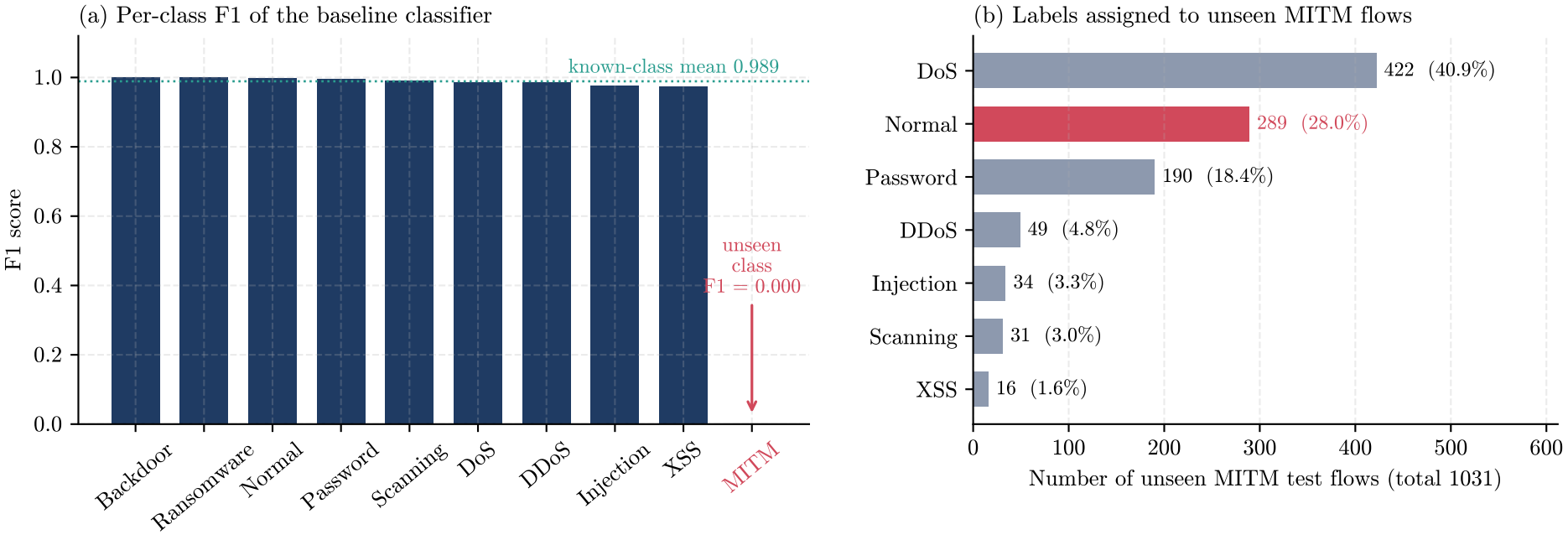
(a) Per-class F1 of the baseline classifier
F1 score
1.0
0.8
0.6
0.4
0.2
0.0
known-class mean 0.989
unseen
class
F1 = 0.000
Backdoor
Ransomware
Normal
Password
Scanning
DoS
DDoS
Injection
XSS
MITM
(b) Labels assigned to unseen MITM flows
DoS
Normal
Password
DDoS
Injection
Scanning
XSS
422 (40.9%)
289 (28.0%)
190 (18.4%)
49 (4.8%)
34 (3.3%)
31 (3.0%)
16 (1.6%)
0
100
200
300
400
500
600
Number of unseen MITM test flows (total 1031)